**JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru**

**DEPARTMENT OF COMPUTER SCIENCE AND IT**

**MCA PROGRAMME**

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# EXP: 03 - QUERY DATA FROM AZURE DATA LAKE STORAGE USING AZURE SYNAPSE ANALYTICS

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SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

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# 1 Aim

To query data stored in Azure Data Lake Storage Gen2 from Azure Synapse Analytics, using the built-in serverless SQL pool, and to inspect the query results.

# 2 Objective and Theory

The objectives are to store a dataset in ADLS Gen2, to query it from Synapse without first loading it into a database, to inspect the returned rows, and to monitor the query. The dataset used is titanic.csv, a structured comma separated file, stored in container rawdata in storage account calablake141. In Synapse serverless SQL, OPENROWSET reads a file directly from storage and presents it as a table, so the file can be queried with ordinary SQL [1]. The same OPENROWSET approach was used again in Experiment 08, over student_performance.csv in the same container.

The file queried here is structured data, a CSV with a header row and fixed columns. No unstructured file, such as a free text file or an image, was queried in this work, and no unstructured query result is claimed in this report.

# 3 Architecture Flow

Data flows from ADLS Gen2 to the Synapse serverless SQL pool, through an OPENROWSET query, to query results and a chart, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 03

| Component | Resource name | Role |
|---|---|---|
| Storage | calablake141 | ADLS Gen2 storage, Hierarchical Namespace enabled, container rawdata. |
| Dataset | rawdata/titanic.csv | Source dataset, 58.89 KiB, Block blob, first row as header. |
| Synapse Workspace | calab-synapse-141 | Synapse workspace, Central India, storage calablake141. |
| SQL Pool | Built-in serverless SQL pool | Serverless pool used for the OPENROWSET query, no dedicated pool. |

*Source: Azure Portal and Synapse Studio screenshots in Section 6, and Experiment 01 report.*

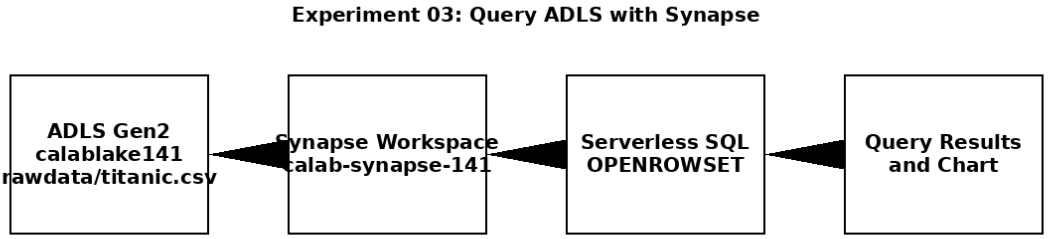



Figure 1.1 Experiment 03 flow

*ADLS Gen2 file to Synapse serverless SQL OPENROWSET query to results and chart.*

# 4 Procedure

**Step 1 (Storage setup).** In storage account calablake141, create Blob container rawdata. The storage account configuration is shown in Screenshot 1.

**Step 2 (Upload dataset).** Upload source file titanic.csv to container rawdata. The container view shows titanic.csv, 58.89 KiB, Block blob, as shown in Screenshot 2.

**Step 3 (Open Synapse Studio).** Open Synapse Studio for workspace calab-synapse-141 and open a SQL script against the built-in serverless SQL pool.

**Step 4 (Run OPENROWSET query).** Run a SQL script using OPENROWSET against https://calablake141.dfs.core.windows.net/rawdata/titanic.csv, with schema mapping for the columns, as shown in Screenshot 3. The query file is in code/synapse_openrowset_query.sql.

**Step 5 (Inspect results).** Inspect the execution output table, which displays queried rows with columns PassengerId, Survived, Pclass, Name, Sex, Age, Fare, Cabin, and Embarked, as shown in Screenshot 4.

**Step 6 (Chart and monitor).** View the query execution results grid with chart visualization, as shown in Screenshot 5, and check the SQL request monitoring log, which shows completed SQL requests and 10.00 MB data processed, as shown in Screenshot 6.

# 5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_03 . The repository contains the README, architecture diagram, the OPENROWSET query, and the Synapse screenshots copied from Experiment 01, where this work was first recorded. The screenshots are the original Experiment 01 files, Step_03, Step_04, Step_11, Step_12, Step_13, and Step_16, kept under their original names so the source can be traced. Table 1.2 records the query.

Table 1.2 Query record, Experiment 03

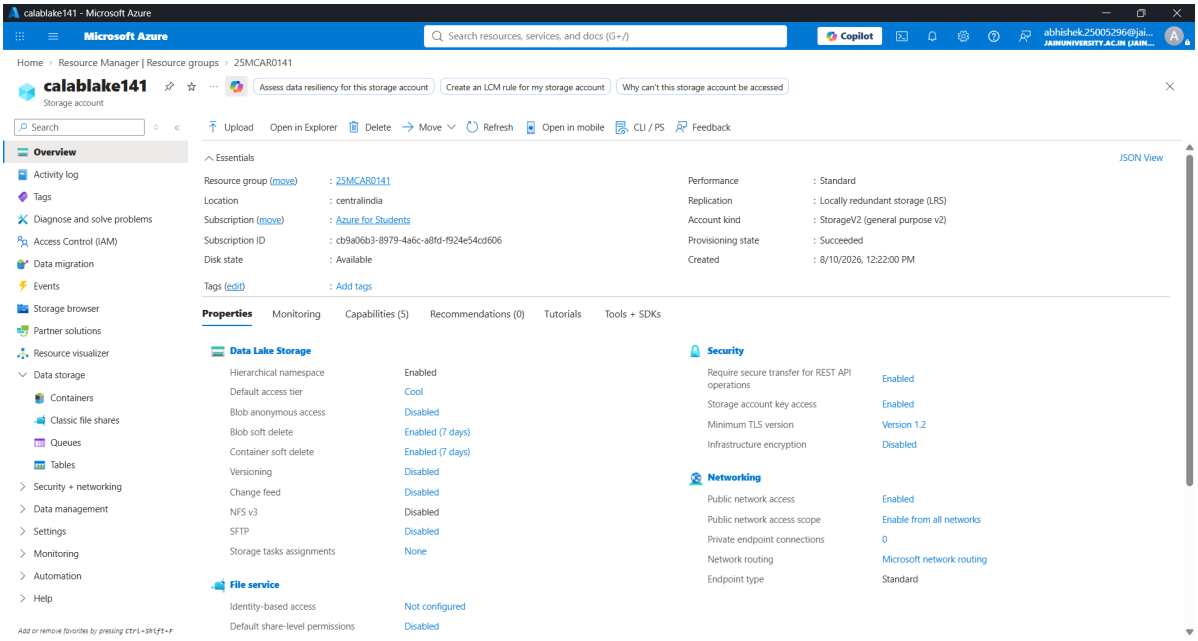
| Item | Value |
|---|---|
| Source file | https://calablake141.dfs.core.windows.net/rawdata/titanic.csv, 58.89 KiB |
| Query method | OPENROWSET, serverless SQL pool, CSV format, first row as header |
| Columns returned | PassengerId, Survived, Pclass, Name, Sex, Age, Fare, Cabin, Embarked |
| Data processed | 10.00 MB, from the SQL request monitoring log |
| Unstructured data | Not queried in this work, no unstructured result claimed |

*Source: Synapse Studio screenshots in Section 6, and Experiment 01 report.*

Result. The titanic.csv file in ADLS Gen2 was queried from Synapse serverless SQL with OPENROWSET, and the returned rows were inspected in the results grid and chart. The query ran, and the monitoring log shows 10.00 MB data processed.

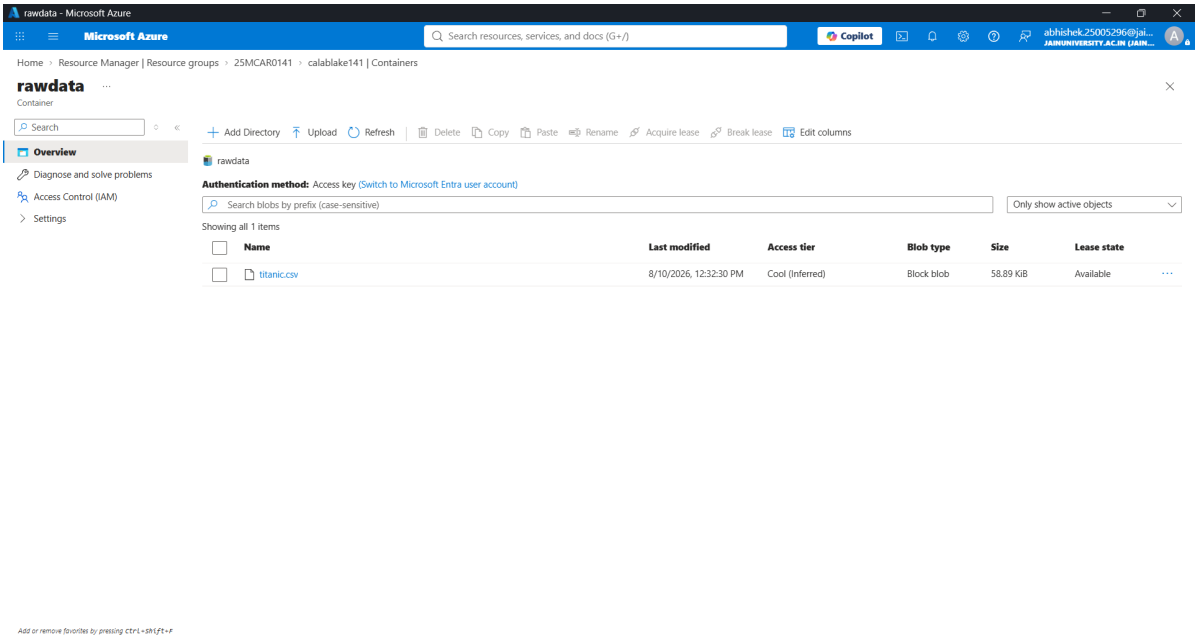
# 6 Screenshots

The screenshots in this section are the original Experiment 01 screenshots, copied without change, showing the ADLS query work that was first recorded under Experiment 01. They are real Azure Portal and Synapse Studio page captures, 1920 by 1020 pixels, embedded at landscape aspect, 6.2 by 3.29 inches. The full Experiment 01 report remains at https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_01 .



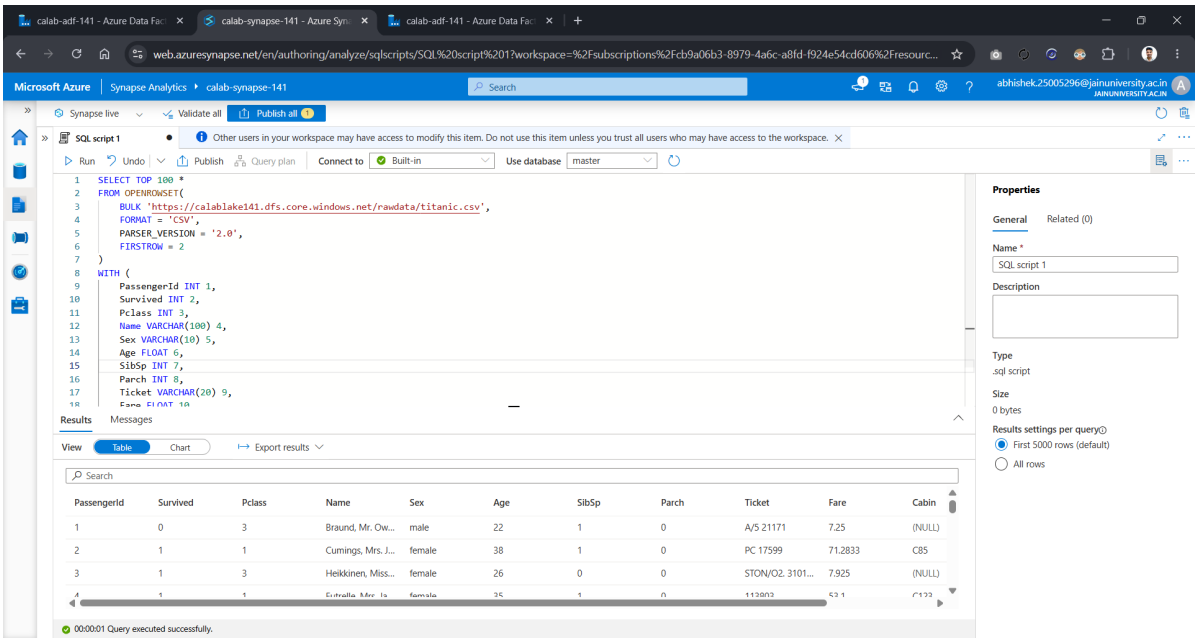

Screenshot 1 Storage Account calablake141 configuration

*Storage Account properties showing StorageV2, Hierarchical Namespace enabled for Data Lake Storage Gen2 capability.*



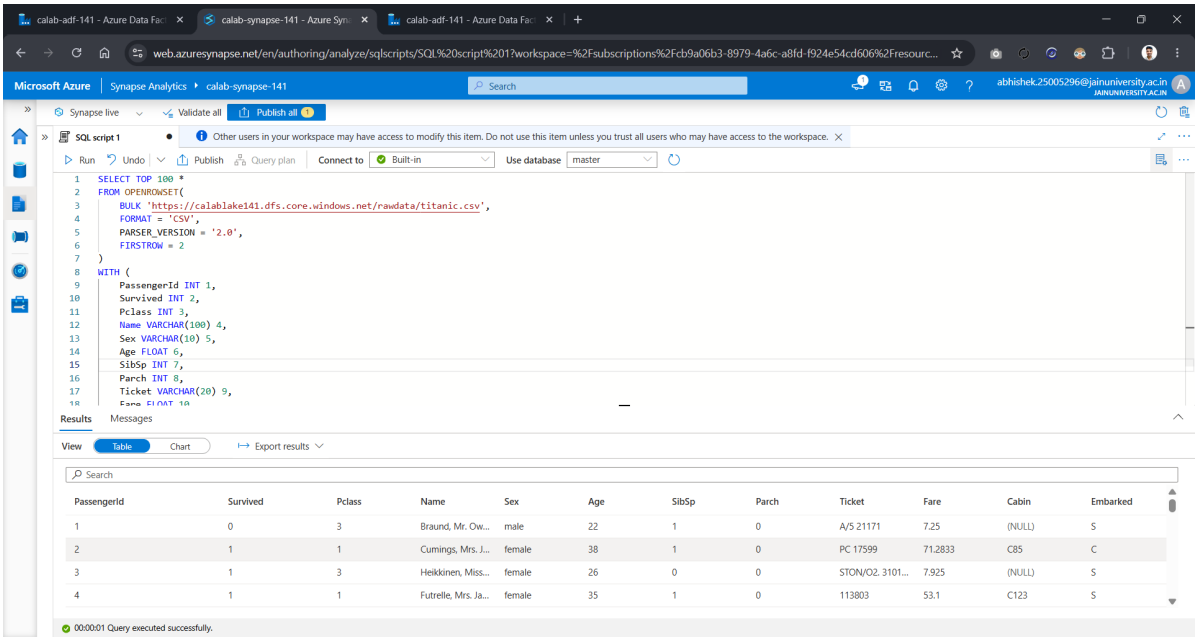

Screenshot 2 Raw Data Container rawdata and source file

*Container rawdata holding the source dataset titanic.csv, 58.89 KiB, Block blob.*



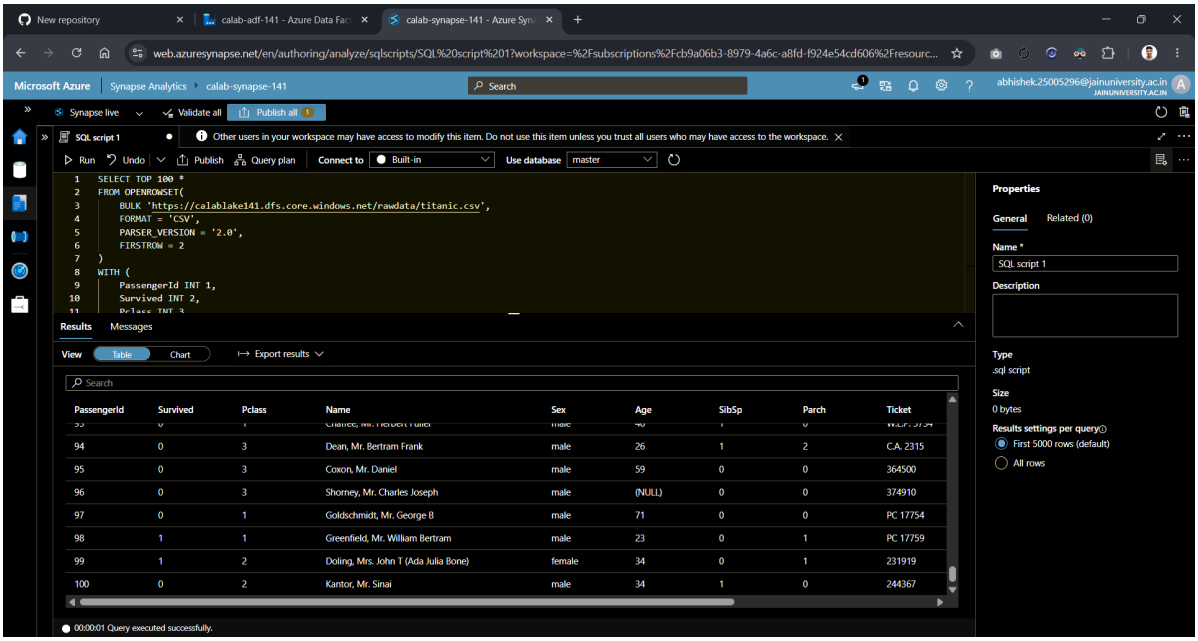

Screenshot 3 Synapse Serverless SQL query definition

*SQL Script executing OPENROWSET against https://calablake141.dfs.core.windows.net/rawdata/titanic.csv with schema mapping.*



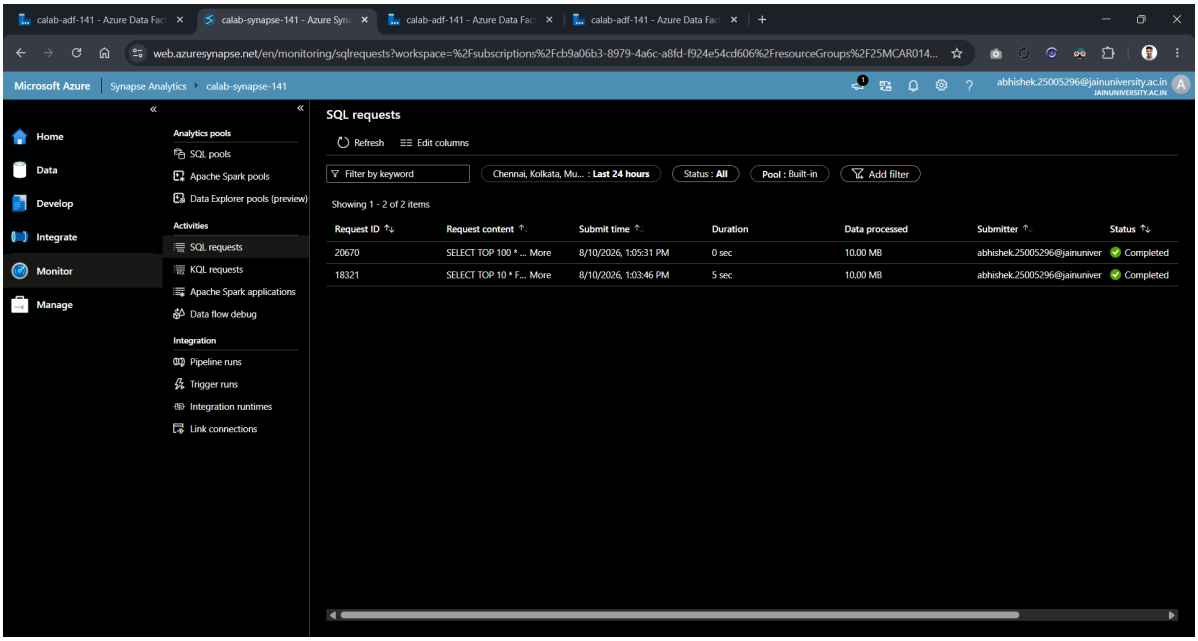

Screenshot 4 Synapse query results and data inspection

*Execution output table displaying queried rows, PassengerId, Survived, Pclass, Name, Sex, Age, Fare, Cabin, Embarked.*




Screenshot 5 Detailed SQL script execution and chart analysis

*Query execution results grid displaying records for Passengers with chart visualization.*




Screenshot 6 Synapse SQL request monitoring log

*Monitoring view showing completed SQL requests, submit times, processing duration, and 10.00 MB data processed.*

Cost note. No hourly pool charge applies in this experiment, because no dedicated SQL pool was created. The serverless SQL query is charged per query on data processed, and the monitoring log shows 10.00 MB data processed.

# References

[1] Microsoft, "Azure Synapse Analytics documentation," Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/synapse-analytics/

[2] Microsoft, "Azure Data Lake Storage Gen2 documentation," Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-introduction